

PREDICTIVE MODELING · MODULE REVIEW

Predictive models: from data to decisions

A visual, intuitive review without fear of math — because understanding comes before memorizing.

DATA

MODELS

METRICS

Hi! I am Maria Eduarda 🖐️



Today, my goal is to translate machine learning concepts into **questions, stories, and decisions** that make sense.

No “black box”

We will understand the reasoning behind the algorithms.

Room to make mistakes

Experimentation is an essential part of data science.

From raw data to knowledge



Data

Explore, clean, and transform.



Prediction

Classification and regression.



Evaluation

Metrics and confusion matrix.



Generalization

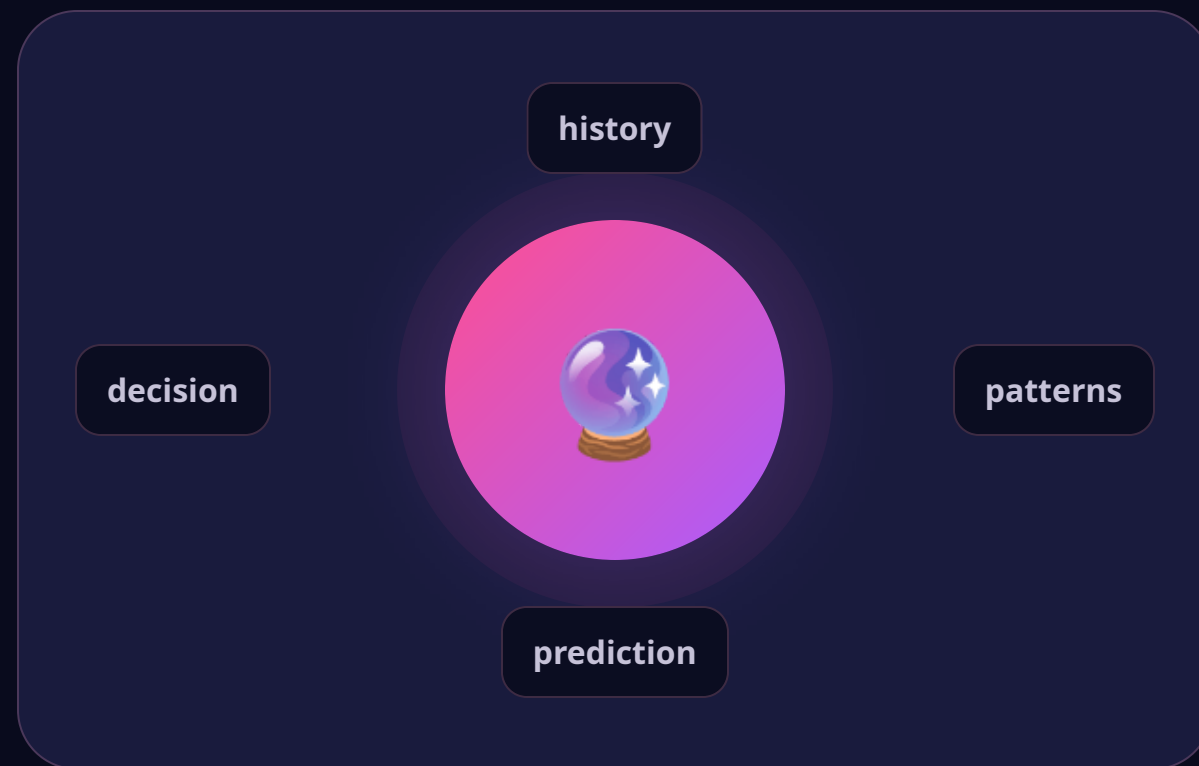
Avoid memorizing examples.



Patterns

Recommendation and clustering.

What is a predictive model?



It is a mathematical rule that learns relationships from past examples to estimate something still unknown.

Like a weather forecaster

She observes today's signals to estimate tomorrow's weather.

Like a recipe

It combines input ingredients to produce an outcome.

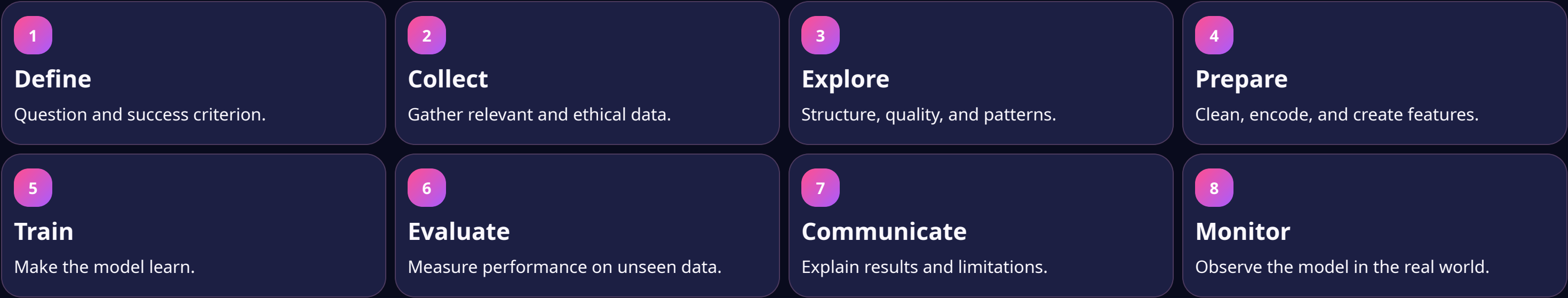
It is not a crystal ball

Predictions carry uncertainty and depend on data quality.

Examples

Fraud, price, demand, risk, churn, and maintenance.

Data science is a cycle



↻ New evidence sends us back to improve earlier decisions.

Tools: each one has a role

Jupyter

Laboratory

Combines code, text, and charts in cells.

Pandas

Smart spreadsheet

Reads, filters, groups, and cleans tables.

NumPy

Numerical engine

Arrays and fast calculations.

Matplotlib

Seaborn

Lenses

Turn numbers into visualizations.

scikit-learn

ML workshop

Prepares, trains, and evaluates models.

Rows, features, and target

ID	age	income	city	purchases	churned?
01	22	3200	SP	4	no
02	35	5100	RJ	1	yes
03	29	?	SP	7	no

row = one observation

features = clues

target = answer to predict

X: inputs
The columns the model can use.

y: output
The column we want to estimate in supervised learning.

Cleaning means making justified decisions

?

Missing values

Remove, fill with median/mode, or represent "unknown"?

×2

Duplicates

Is it a collection error or two truly identical observations?

ABC

Categories

Standardize spelling and convert text into a numerical representation.

↔

Scales and outliers

Compare magnitudes and investigate extremes before removing them.

Analogy: before cooking, **we wash, cut, and measure the ingredients.**

Exploring means interviewing the data

Every chart should answer a question — not merely “look pretty.”

What is the shape?

Number of rows, columns, and types.

What is the quality?

Missing values, duplicates, and impossible values.

How is it distributed?

Mean, median, range, and outliers.

What is related?

Compare groups and variables.



Feature engineering: give meaning to the clues

date of birth

individual purchases

free-text address

two related columns



age

frequency and average order value

city / region

ratio or difference

Good feature

Represents the phenomenon and exists at prediction time.

Data leakage

When a feature reveals information the model would not have in the real world.

With answers or without answers?



SUPERVISED

Training with an answer key

Examples have a known target; the model learns to predict that answer.

- Classification
- Regression
- E.g., spam, price, risk



UNSUPERVISED

Exploration without an answer key

There is no target; we look for structure, proximity, and hidden patterns.

- Clustering
- Dimensionality reduction
- Anomalies and associations

Category or quantity?

CLASSIFICATION

Which class?

Predicts a category: yes/no, fraud/normal, species A/B/C.

Output

Class label or probability.

REGRESSION

How much?

Predicts a continuous value: price, temperature, demand, or time.

Output

A number on a continuous scale.

“Will this customer churn?”

Binary classification.

“How much will they spend?”

Regression.

The model must take a test too

“Memorizing the exercise list does not guarantee learning for the test.”

Splitting the data measures **generalization**: performance on new cases.

Training
learns parameters

70%

Validation
selects and tunes

15%

Test
evaluates once at the end

15%

Cross-validation

Repeats training and validation across folds for a more stable estimate.

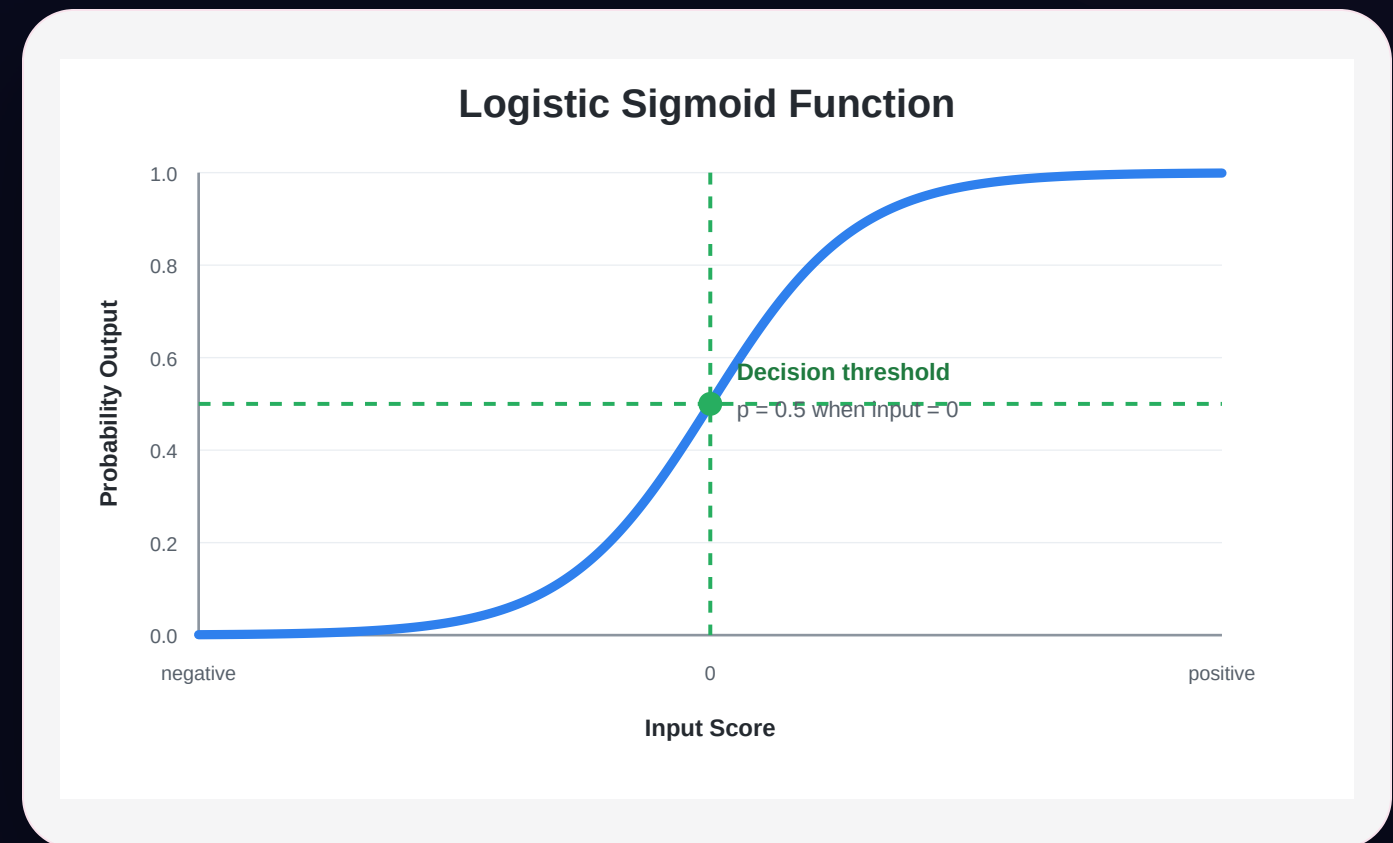
Logistic Regression: a probability gate

It combines evidence and uses the sigmoid curve to turn the result into a probability between 0 and 1.

- 1 Combines features with weights.
- 2 Applies the sigmoid function.
- 3 Compares against a threshold, usually 0.5.
- 4 Returns a probability and a class.

Analogy

A gate that gradually opens as the evidence grows.



A tree asks. A forest votes.

Decision Tree

Chains “if/then” questions. It is explainable, but may memorize details.

Random Forest

Trains many trees on different samples and combines their votes.

Hyperparameters

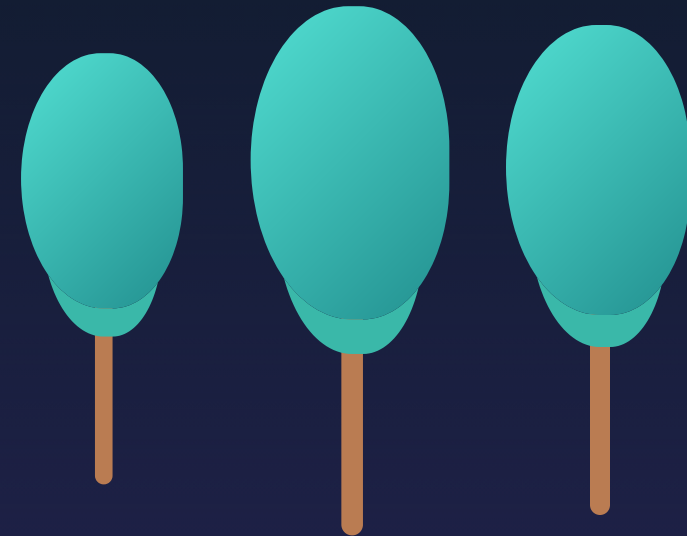
Depth, number of trees, and minimum samples per leaf.

Feature importance

Shows which variables contributed most to decisions.

Analogy: **a panel of experts** is usually more stable than a single opinion.

majority vote



KNN: ask the nearby neighbors

To predict a new example, KNN consults the **K most similar examples** and combines their answers.

- 1 Computes distance to the known data.
- 2 Selects the K nearest neighbors.
- 3 Uses voting for classification or an average for regression.

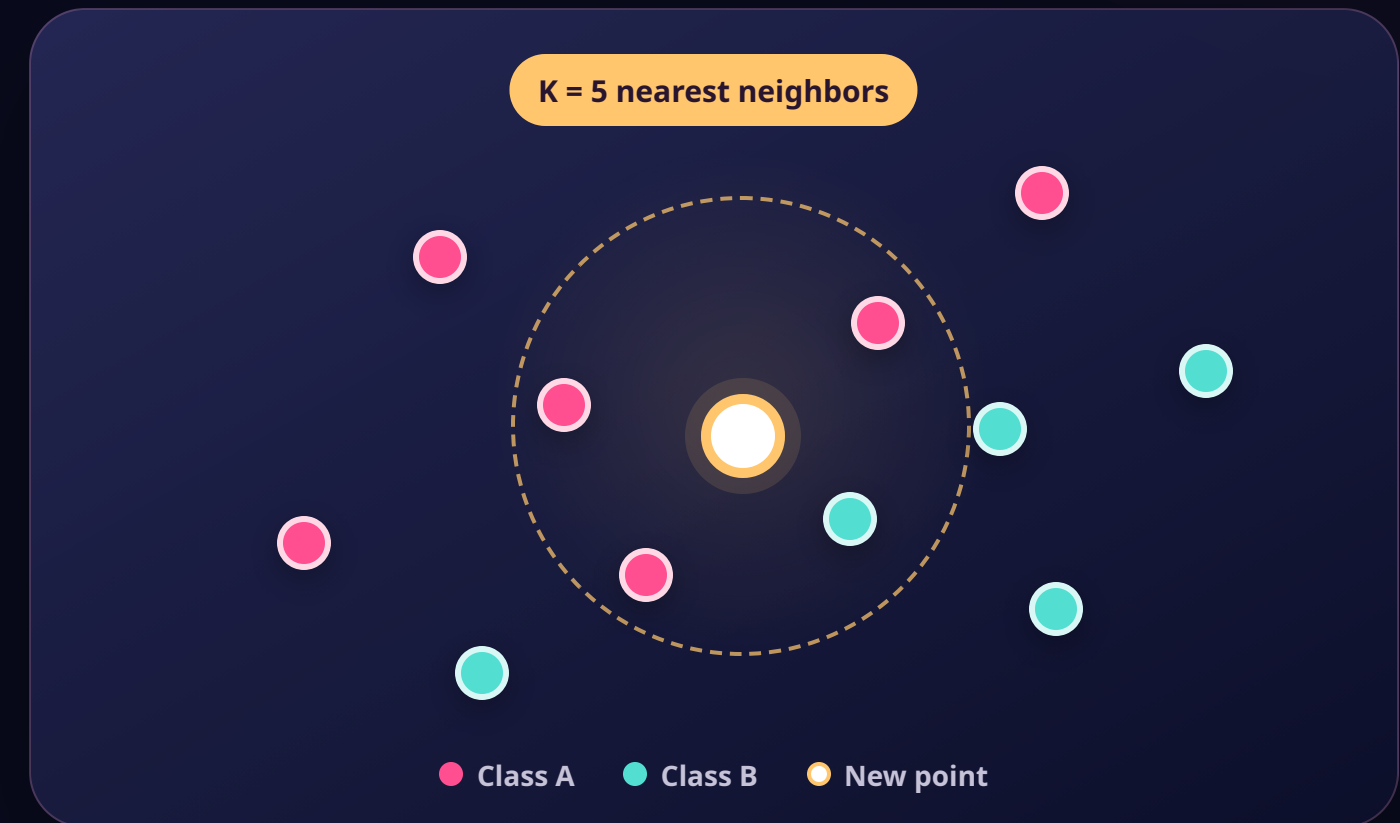
Small K

Sensitive to noise.

Large K

May oversmooth.

Essential: standardize features, because distance depends on scale.



Naive Bayes: update a belief with evidence

It starts with an initial probability and asks: **how likely are these clues in each class?**

- 1 Defines the initial chance of each class — the prior.
- 2 Measures the strength of the evidence — the likelihood.
- 3 Updates the probabilities and chooses the largest posterior.

Why “naive”?

It assumes features are independent within each class. This is a simplification, but it works very well for many text tasks.

Fast and efficient for spam, sentiment, and document classification.

EXAMPLE: IS THIS EMAIL SPAM?

“free”

“prize”

many links

$\text{posterior} \propto \text{evidence} \times \text{prior}$



Spam



88%

Not spam



12%

Analogy

A detective updates her hypothesis whenever she finds a new clue.

First, count correct and incorrect predictions

	Predicted no	Predicted yes
Actual no	TN correct negative	FP false alarm
Actual yes	FN missed positive	TP correct positive

The diagonal contains correct predictions. The two error types are off the diagonal.

False Positive

The model raised an alert, but the event did not happen.

False Negative

The event happened, but the model did not detect it.

Essential question: which error is more costly for the problem?

METRICS

Accuracy

$$(TP + TN) / \text{all}$$

How many predictions were correct overall?

Useful with balanced classes.

Precision

$$TP / (TP + FP)$$

Of the alerts, how many were correct?

Controls false alarms.

Recall

$$TP / (TP + FN)$$

Of the actual positives, how many did we find?

Controls missed cases.

F1

harmonic mean of P and R

How can we balance Precision and Recall?

Useful with class imbalance.

ROC-AUC

$$0.5 \rightarrow 1.0$$

Does the model rank positives above negatives?

Evaluates many thresholds.

Model Review

Accuracy can mislead

If 99% of transactions are normal, always predicting “normal” gives 99% accuracy while detecting zero fraud

The metric comes from the cost of error

How do we measure error size?

Metric	Idea	How to interpret	Better direction
MAE	Mean absolute error	Typical error in the target's units	Lower
MSE	Mean squared error	Strongly penalizes large errors	Lower
RMSE	Square root of MSE	Penalizes large errors but returns to the original unit	Lower
R ²	Explained variance	1 is perfect; 0 matches the mean baseline; it may be negative	Higher

MAE is a standard ruler

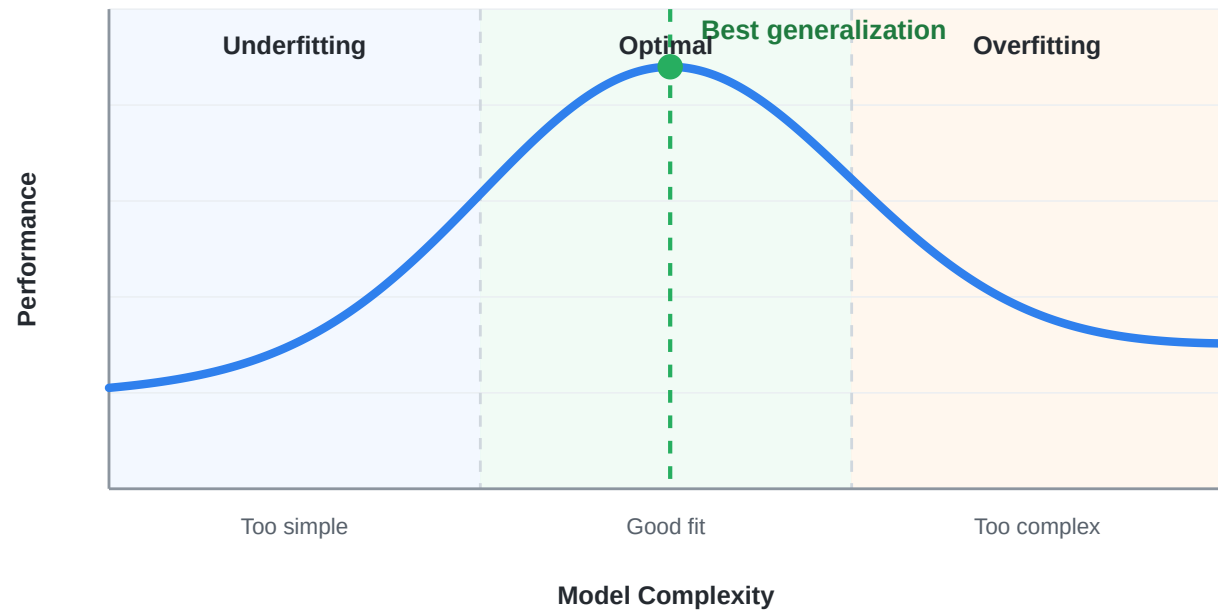
Each error has proportional weight.

RMSE is a sensitive ruler

Very large errors receive more weight.

Overfitting: when the model memorizes

Model Complexity and Performance



"I memorized the answer sheet, but the test brought new questions."

Underfitting

Model is too simple: poor on both training and validation.

Good generalization

Learns the pattern and works on new data.

Overfitting

Excellent training score, worse validation score: it learned noise and details.

Countermeasures: more data, cross-validation, regularization, fewer features, a smaller tree, and a simple baseline.

How do we choose and improve a model?

1 · Baseline

Start with a rule or simple model.

2 · Fair comparison

Same split, features, and metric.

3 · Cross-validation

Observe the mean and variation across folds.

4 · Hyperparameters

Tune through search without looking at the test set.

5 · Errors

Investigate where and for whom the model fails.

Performance

Does the metric match the objective?

Explainability

Can we justify decisions?

Cost

Are time, memory, and maintenance worth the gain?

The “best” model is the one that solves the problem with **evidence and responsibility.**

Recommendation: “you may like this”

It is like a friend who knows your taste — and also knows people similar to you.

Collaborative filtering

What did people with similar behavior like?

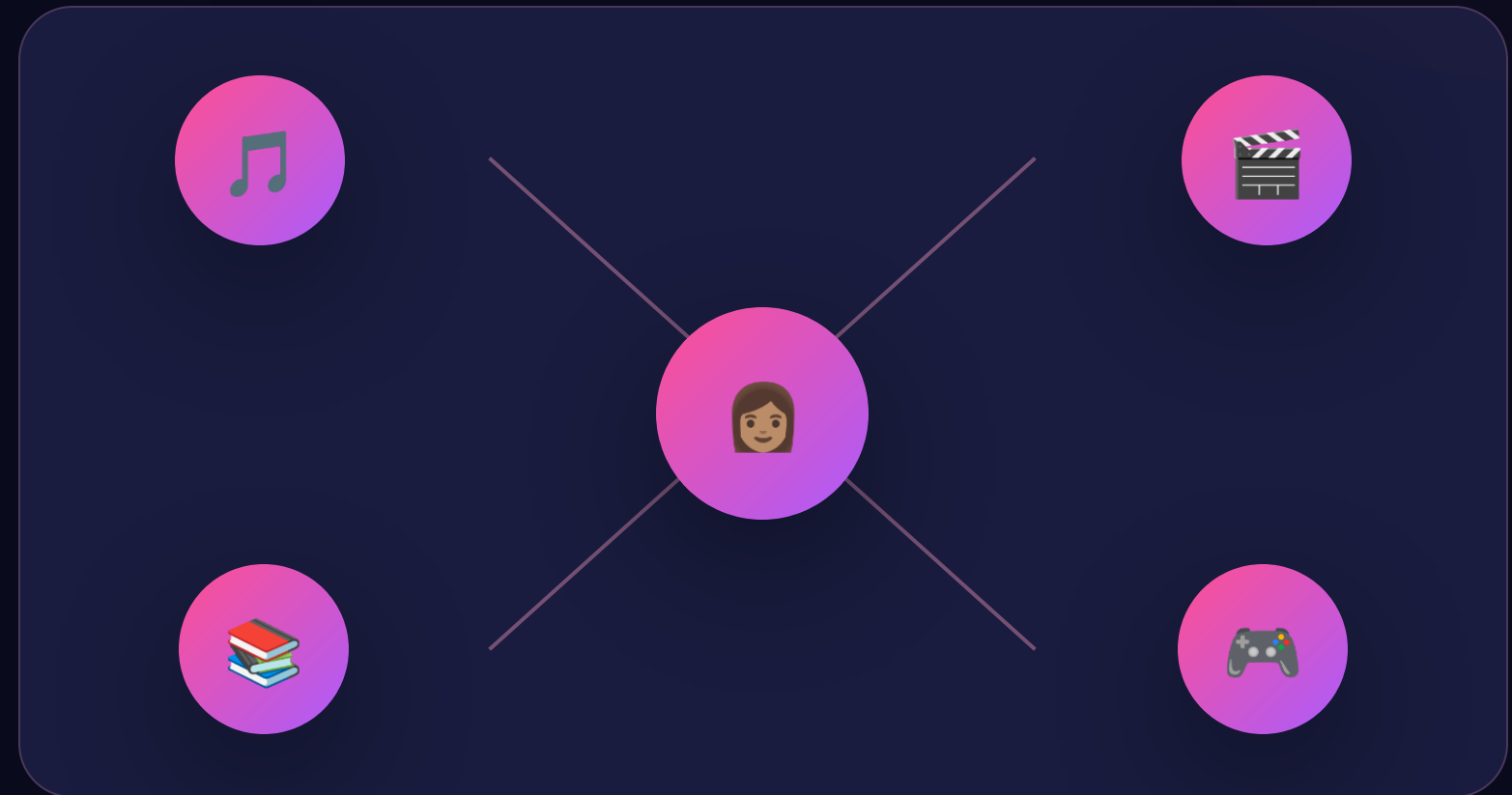
Content-based

Which items resemble the ones you already liked?

Hybrid

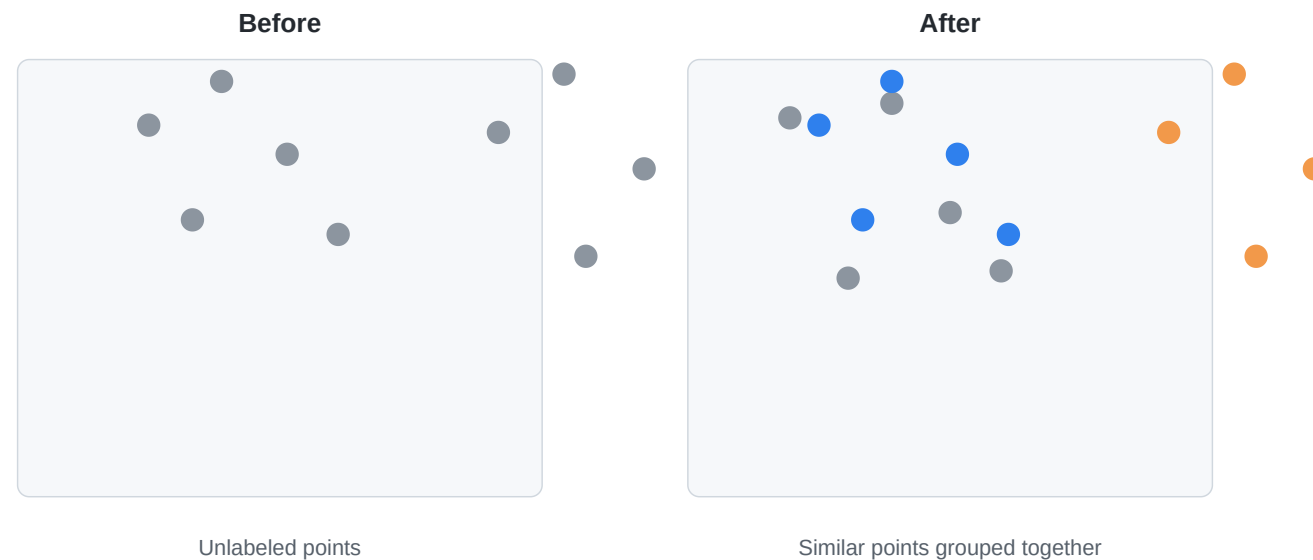
Combines behavior and item characteristics.

Common metrics: Precision@K, Recall@K, NDCG, MAP, and rating error.



Clustering: find groups without labels

Before and After Clustering



Analogy: organize a closet without labels by grouping similar-looking items.

Within a cluster

Items should be similar.

Between clusters

Groups should be different.

Applications

Segmentation, topics, compression, and pattern discovery.

K-Means: bring each point closer to a center

Imagine placing **K meeting points** to minimize everyone's travel distance.

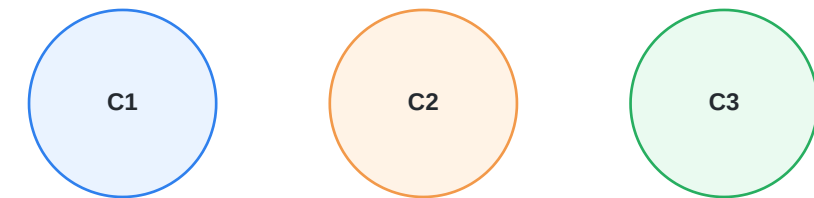
- 1 Choose K initial centroids.
- 2 Assign each point to its nearest center.
- 3 Move each center to the group mean.
- 4 Repeat until stable.

Inertia

Sum of squared distances to centroids. Lower means more compact groups.

Feature scaling is essential: a large scale may dominate distance.

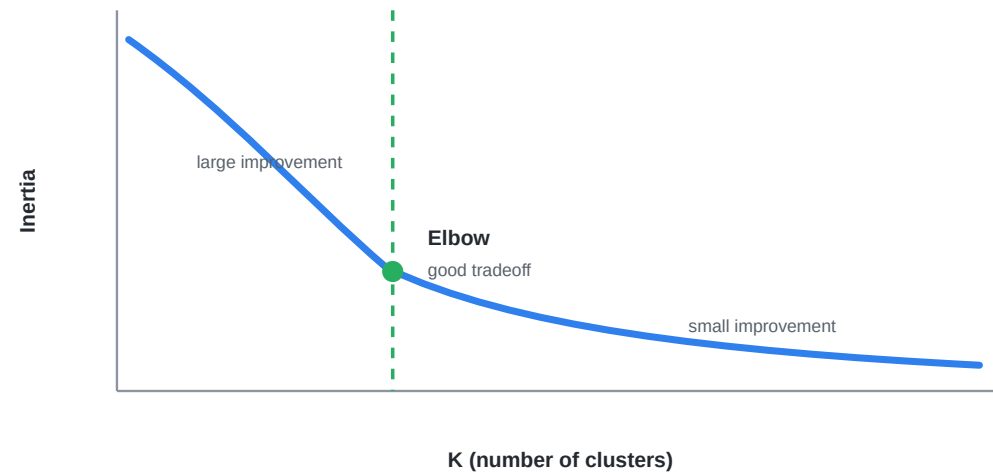
K-Means Algorithm



Centroids shift until point assignments stop changing.

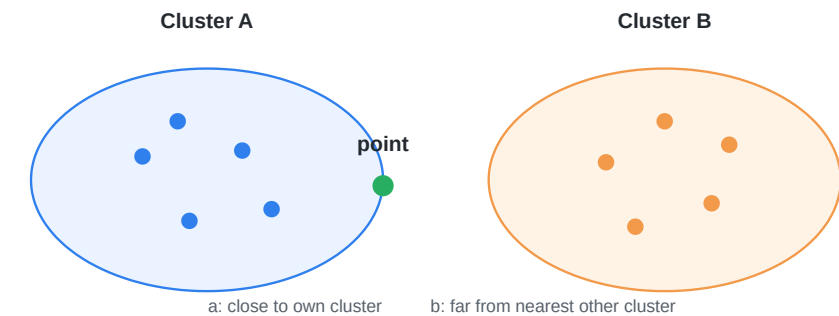
Number of groups: use two sources of evidence

Elbow Method



Elbow: when do the gains begin to shrink?

Silhouette Score Intuition



Silhouette: internal cohesion + separation

Elbow

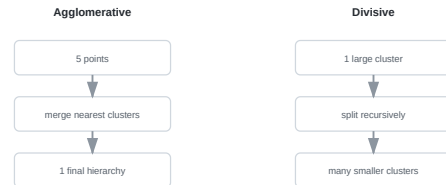
Inertia always decreases; we look for the curve's "elbow."

Silhouette: -1 to +1

Near 1 means well grouped; 0 means a boundary; negative suggests a poor assignment.

Group shape and the meaning of distance matter

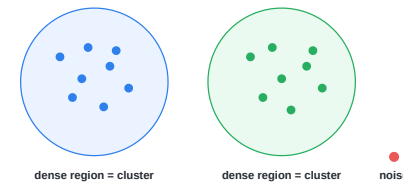
Hierarchical Clustering



Hierarchical

Builds a tree of groups. The dendrogram shows the distance at which groups merge.

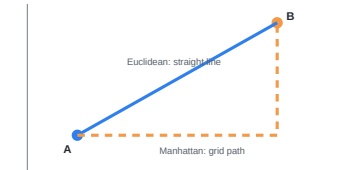
DBSCAN: Density and Noise



DBSCAN

Finds dense regions, irregular shapes, and noise points. It uses *eps* and *minPts*.

Euclidean vs Manhattan Distance



Distances

Euclidean: straight line. Manhattan: city blocks. Cosine: direction. Jaccard: overlap.

There is no neutral distance

The choice defines what “similar” means to the algorithm.

NEXT STEP · PRACTICE

Kaggle is the laboratory. You are the scientists.

Use notebooks and challenges to turn questions into experiments: explore, prepare, train, evaluate, explain, and try again.

Thank you! ❤️

Keep creating and transforming technology — with curiosity, courage, and collaboration.